

 OR-AS Operations Research Applications and Solutions	Case Name: Drainage System Project Owner: Eder Omar Vilca Carrillo	Sector	Construction
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
			One of nine std. scenarios
Processed by	Amin Azimi		User defined distributions
Date	2025	Project Control	Automatic tracking
File Name	C2025-14		Tracking based on user input

1. Project description

Project authenticity

Repair of the pluvial (stormwater) sewer system to improve flood management and water flow in the affected area.

The project involves preliminary works, drainage structure maintenance, and infrastructure rehabilitation.

The project consists of activity and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	82	Serial/Parallel (SP)	20%
Planned Duration (PD)	354 days	Activity Distribution (AD)	35%
Budget At Completion (BAC)	32,341.63€	Length of Arcs (LA)	0%
Renewable Resources	6 Types	Topological Float (TF)	43%
Consumable Resources	0 Types		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	12.48	9.37	1.89
CRI-rho	12.37	9.26	1.78
CRI-tau	9.29	7.00	1.31

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	14.77	8.33	5.22
CRI-rho	14.77	8.33	5.22
CRI-tau	14.77	8.33	5.22

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	13.59	33.51	2.18
SI	21.45	31.48	2.04
SSI	3.33	10.67	3.91
CRI-r	10.20	10.53	2.64
CRI-rho	10.10	10.25	2.68
CRI-tau	12.23	7.62	0.65

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	171.30	-101.70
PV - SPI	79.44	72.16
PV - SCI	79.74	71.96
ED - 1	7839.00	7839.00
ED - SPI	79.44	72.16
ED - SCI	79.03	71.81
ES - 1	14.03	-11.42
ES - SPI(t)	29.21	11.56
ES - SCI(t)	29.21	11.57

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0.57	0.00
CPI	0.60	0.00
SPI	14.50	14.50
SPI(t)	13.97	13.97
SCI	14.50	14.50
SCI(t)	13.98	13.98
0.8 CPI + 0.2 SPI	13.93	13.93
0.8 CPI + 0.2 SPI(t)	11.82	11.82

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over **three** tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	0.00	-7211.18	-124.00	1	0.78	0.96	0.73
std dev	0.00	6621.62	27.22	1	0.20	0.01	0.28
final	0.00	0.00	-99.00	1	1.00	0.97	1.00

2.3.3.2. Time forecasting

PD	354 days	Real Duration	388 days	Delay	9.6%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	432.33	72.78	17.44	-11.43
PV - SPI	475.67	119.63	28.61	-22.59
PV - SCI	475.67	119.63	28.61	-22.59
ED - 1	439.00	66.46	16.92	-13.14
ED - SPI	482.67	113.18	28.18	-24.40
ED - SCI	482.67	113.18	28.18	-24.40
ES - 1	369.00	3.61	4.90	4.90
ES - SPI(t)	369.33	3.51	4.81	4.81
ES - SCI(t)	369.33	3.51	4.81	4.81

2.3.3.3. Cost forecasting

BAC	32,341.63€	Real Cost	32,341.63€	On Budget	0%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	32341.64	0.00	0.00	0.00
CPI	32341.64	0.00	0.00	0.00
SPI	36307.78	4444.49	12.26	-12.26
SPI(t)	32694.71	309.27	1.09	-1.09
SCI	36307.78	4444.49	12.26	-12.26
SCI(t)	32694.71	309.27	1.09	-1.09
0.8 CPI + 0.2 SPI	32883.17	571.97	1.67	-1.67
0.8 CPI + 0.2 SPI(t)	32409.56	59.56	0.21	-0.21